SHEET # 1

1.1 Define the following items: Signal, system, signal processing. (State one example for each).

1.2 Classify the different types of signal processing.

1.3 Classify the different signal types.

1.4 Explain the basic three operations performed in A/D converters.

1.5 State and explain the sampling theorem.

1.6 Determine which of the following sinusoid signals are periodic, and then compute their fundamental period.

(a) $x_{a(t)}$ = 3cos(5t +π/6) (b) $x_{a(t)}$ =3cos(100πt)

(c) x(n) = cos[π(30n/105)] (d) x(n) = 2exp.[j(n/6-π)]

1.7 Consider the following analog sinusoid signal: $x_{a(t)}$= 3sin (100πt)

(a) Sketch the signal for [0, 30] ms.

(b) The signal $x_{a(t)}$ is sampled with sampling rate F_S=300 samples per second.

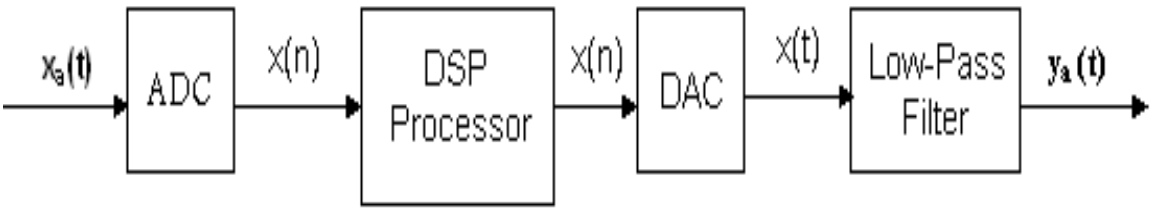
Determine the frequency of the discrete-time signal and show that it is periodic.

(c) Compute the sample values in one period of x (n). Sketch x (n) on the same diagram with $x_{a(t)}$. What is the period of this discrete-time signal, x (n).?

(d) Can you find a sampling rate F_S such that the signal X(n) reaches its peak value 3 ? What is the minimum F_S suitable for this task?

1.8 Consider a simple signal processing system as shown in the following figure. The sampling periods of the A/D and D/A converters are T_1=5 ms and T_2=1 ms respectively. Determine the output y_a (t) if the input is given by:

$x_{a(t)}$ = 3 cos (100πt) + 2 sin (250πt) (t in second)



Assignment # 1

A1.1 An analog signal contains up to 10 KHz.

a- What is the range of sampling frequencies allows exact reconstruction of this signal from its samples?

b- Suppose that we sample this signal with sampling frequency F_S = 8 KHz. Is it possible to reconstruct this signal without distortion?

c- For the same sampling frequency given in (b), examine what was happen if the frequency of this signal is 24 KHz. Cheek the periodicity of the obtained x (n) and if it is periodic find its fundamental period. Find the first four samples of x (n).

A1.2 Determine, which of the following sinusoid signals, is periodic, and then compute their fundamental period.

a- $x(n) = \cos(3n\pi)$ b- $x(n)= 3\cos(5n +\pi/6)$ c- $x(n)= \sin(3n)$

A1.3 Consider the analog signal $x_a(t) = 3 \cos 100\pi t$

a- Determine the minimum sampling rate required to avoid aliasing.

b- Assume that this signal is sampled at the rate $F_s = 200$ Hz. What is the discrete-time signal obtained after sampling, x(n)? Find the first four samples.

c- Plot $x_a(t)$ and x(n) obtained in (b) on the same graph.